

QM. Imp.

$$\rightarrow \Delta E \cdot \Delta t \geq \frac{\hbar}{2}$$

$$\rightarrow \int_a^b |\Psi(x,t)|^2 dx \quad \text{prob. of finding a particle between point a and b.}$$

$\rightarrow$ measurement - an active interaction that forces the particle to collapse from a superposition of possibilities into a single definite state.

$\hookrightarrow$ then we get a probability.

$$\rightarrow \Psi = Ae^{-(i/\hbar)(Et - px)} = Ae^{i\frac{(Et - px)}{\hbar}}$$

$\hookrightarrow$ equation of motion of free particle. with energy E , momentum p , moving in (x) direction.

$\rightarrow$ In QM Ψ and $e^{i\theta}\Psi$ are equivalent as the physical significance (probability) does not change.

$$|\Psi|^2 = \Psi^* \Psi = (e^{i\theta}\Psi)^*(e^{i\theta}\Psi)$$

$e^{i\theta} \rightarrow$ global phase factor, has no physical significance.

$$\rightarrow \text{TISE} \Rightarrow \Psi(x,t) = \Psi(x) \cdot e^{-\frac{iEt}{\hbar}}$$

$\rightarrow$ Any well behaved function
* fig, Ψ etc. vanish at $\pm\infty$.

$E \rightarrow$ Energy
of that
eigenstate.